# Erratum: Maximum time delay achievable on propagation through a slow-light medium [Phys. Rev. A 71, 023801 (2005)]

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(Received 12 August 2005; published 4 November 2005)

DOI: 10.1103/PhysRevA.72.059903 PACS number(s): 42.50.Gy, 42.65.Tg, 99.10.Cd

We are grateful to Bruno Macke of the University of Lille for calling to our attention several errors in the equations of our paper [1]. These errors do not appear in the final results of our calculation nor do they influence the conclusions of our work, but they could present a problem for someone attempting to reproduce the details of our calculation. These errors are as follows.

The last term of Eq. (4) should read $+f\delta^2/\gamma^2$ instead of $-f\delta^2/\gamma^2$. Also Eq. (12) should read

$$A(\delta) = A_0 \exp\left[-\frac{\delta^2 T_0^2}{2} - \frac{f\alpha_0 L}{2}\frac{\delta^2}{\gamma^2} - ikL - \frac{\alpha_0 L}{2}(1-f)\right].$$

Three lines below Eq. (13), the in-line equation should read $L_{\max}=3T_0^2\gamma^2/f\alpha_0$. Fourteen lines below Eq. (15), the in-line equation should read $T_0=50/\gamma$ instead of $T_0=50\gamma$. Also, a factor of $i\gamma_{ba}$ is missing in the numerator of Eq. (16), and the text following this equation should read "where $\gamma_{ba}$ is the coherence dephasing rate of the probe transition, $\Omega_s$ is the Rabi frequency of the strong coupling field which is tuned near the $bc$ transition, $\Delta$ is the detuning of the coupling field from exact resonance, and $\gamma_{ca}$ is the dephasing rate of the ground state coherence. For a dilute medium the refractivity $(n-1)$ and the absorption coefficient $\alpha$ are respectively given by $\frac{1}{2}\,\mathrm{Re}\,\chi^{(1)}$ and $(\omega/c)\mathrm{Im}\,\chi^{(1)}$."

[1] R. W. Boyd, D. J. Gauthier, A. L. Gaeta, and A. E. Willner, Phys. Rev. A **71**, 023801 (2005).

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